AMI DATA ANALYTICS

Analyzer Module Technical Reference Guide

Architecture, Data Flow, Backend Services, Snowflake Integration, and Support Documentation

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Application

AMI Data Analytics Platform

Module

Analyzer

Table of Contents

1. Overview

1.1 Purpose

This document provides a detailed technical reference for the Analyzer module, including frontend components, backend services, API integrations, Oracle stored procedures, Snowflake query execution, business calculations, and troubleshooting guidance.

1.2 Scope

- Meter Analysis
- Coincident / Non-Coincident Peak Reporting
- Graph Framework
- Snowflake Integration
- Export Features

1.3 Major Functional Areas

The Analyzer module provides analytical capabilities for AMI meter and transformer data. The module contains two major functional areas:

1.3.1 Meter Analysis

Used to analyze:

- Demand
- Consumption
- Voltage

for individual meters.

1.3.2 Coincident / Non-Coincident Peak Reporting

Used to:

- Search transformers
- Calculate coincident peaks
- Calculate non-coincident peaks
- Identify contributor meters
- Generate reports

2. System Architecture

2.1 High-Level Architecture

```
Angular UI
↓
Controller APIs
↓
Service Layer
↓
Oracle Stored Procedures
↓
Oracle Query Repository
↓
Snowflake Queries
↓
Snowflake Data
↓
JSON Response
↓
Charts / Grids
```

2.2 Component Architecture

The following matrix defines component dependencies across the Analyzer module.

Component	Depends On
AnalyzerComponent	MeterAnalyzerService, TransformerService, SearchComponent
MeterAnalysisDetailsComponent	MeterAnalyzerService
GraphController	GraphService
GraphService	OracleConnector, SnowflakeConnector

2.3 Data Flow Architecture

End-to-end architecture summary across the module's primary workflows.

Meter Search → Analysis

```
Meter Search
↓
AnalyzerComponent
↓
MeterAnalyzerService
↓
MeterAnalyzerController
```

↓

`Oracle Stored Procedure`

Meter Details → Graphing

`Meter Details`

↓

`MeterAnalysisDetailsComponent`

↓

`GraphController`

↓

`GraphService`

↓

`Oracle Query Repository`

↓

`Snowflake`

Transformer Search

`Transformer Search`

↓

`AnalyzerComponent`

↓

`MeterAnalyzerController`

↓

`Oracle Procedure`

Coincident Popup

`Coincident Popup`

↓

`TransformerService`

↓

`Snowflake`

↓

`Contributor Meter Data`

3. Frontend Architecture

3.1 Primary Component

Page: Analyzer → Meter Analysis

Component	Files
AnalyzerComponent	analyzer.component.ts, analyzer.component.html

3.2 Supporting Components

Component	Responsibility
SearchComponent	Used for Meter Search and Transformer Search
UtilityTypeFilterCellComponent	Grid filter component
PcbFilterCellComponent	Transformer PCB filter

3.3 Frontend Services

- MeterAnalyzerService.service.ts
- TransformerService.service.ts
- MeterService.ts
- SecurityService.ts

4. Backend Architecture

4.1 Controller Layer

Controller	Key Methods	Purpose
MeterAnalyzerController	AnalyzerAllMeterDetails(), SearchAnalyzerMeterDetails(), AnalyzerCoinNonCoinPeakReportSearch()	Handles meter and transformer search, and peak report search requests
GraphController	GetCombinedMeterData()	Retrieves combined chart data via GraphService

4.2 Service Layer

Service	Key Methods	Purpose
MeterAnalyzerService	GetAllAnalyzerMeterDetailbyJson(), SearchAnalyzerMeterDetailsByParam()	Retrieves and searches meter details
GraphService	GetCombinedMeterData(), GetQueryFromOracle(), ReplacePlaceholdersFromJson()	Resolves dynamic queries and retrieves chart data

4.3 Connector Layer

Connector	Key Methods	Purpose
IOracleConnectorODP	—	Oracle database connectivity
SnowflakeConnector	ExecuteSnowflakeJson(), fetchSnowflakeDataASDT(), GetAllDataAsJsonString()	Executes Snowflake queries and returns JSON results

4.4 Backend Code Inventory

Controllers

- MeterAnalyzerController.cs
- GraphController.cs

Services

- MeterAnalyzerService.cs
- GraphService.cs

Connectors

- IOracleConnectorODP
- SnowflakeConnector.cs

5. Query Resolution Framework

5.1 Dynamic Query Framework Overview

▯ **Important Note:** The Analyzer module does not contain hardcoded Snowflake SQL. All Snowflake query definitions are retrieved dynamically from Oracle at runtime.

5.2 Query Resolution Process

```
Query ID
   ↓
GraphService
   ↓
GetQueryFromOracle()
   ↓
V_SP_DA_GET_ALGQUERY
   ↓
Query Version Repository
   ↓
SQL Template
   ↓
ReplacePlaceholdersFromJson()
   ↓
Snowflake SQL
   ↓
ExecuteSnowflakeJson()
```

5.3 Feature-to-Query Mapping

Feature	Query ID
Voltage Analysis	15
Detailed Demand	16
Detailed Consumption	16
Billing Periods	17
Coincident Peak — 15 Min	63
Coincident Peak — Hourly	65
Coincident Peak — Daily	66

5.4 Query ID Inventory

Query ID	Feature	Description
15	Voltage Analysis	Retrieves per-phase voltage readings for a meter

Query ID	Feature	Description
16	Detailed Demand	Retrieves demand metrics for a meter
16	Detailed Consumption	Retrieves consumption metrics for a meter
17	Billing Periods	Retrieves billing period date ranges
63	Coincident / Non-Coincident (15 Min)	Coincident/non-coincident peak query, 15-minute interval
65	Coincident / Non-Coincident (Hourly)	Coincident/non-coincident peak query, hourly interval
66	Coincident / Non-Coincident (Daily)	Coincident/non-coincident peak query, daily aggregated

6. Meter Analysis Module

6.1 Meter Search

6.1.1 Purpose

Allows users to search AMI meters.

6.1.2 Search Criteria

- Meter Number
- Address
- City
- Account Number
- Point ID

6.1.3 Data Flow

Meter Search Flow

```
User enters search criteria
↓
SearchComponent
↓
handleSearch()
↓
AnalyzerComponent
↓
getAnalyzerMetersDetailsBySearchParam()
↓
MeterAnalyzerService
↓
POST SearchAnalyzerMeterDetails
↓
MeterAnalyzerController
↓
MeterAnalyzerService
↓
V_SP_DA_ANALYZER_METER_SEARCH
↓
Oracle
↓
JSON Response
↓
Fusion Grid
```

6.1.4 API Reference

Operation	Method	Endpoint
Load All Meters	GET	/MeterAnalyzer/AnalyzerAllMeterDetails
Search Meters	POST	/MeterAnalyzer/SearchAnalyzerMeterDetails

Search Meters — Request Body

```
{ "Meter_no": "", "Address": "", "City": "", "Account_no": "", "Point_id": "" }
```

6.1.5 Controller and Service Methods

Layer	Methods
Controller (MeterAnalyzerController)	AnalyzerAllMeterDetails(), SearchAnalyzerMeterDetails()
Service (MeterAnalyzerService)	GetAllAnalyzerMeterDetailbyJson(), SearchAnalyzerMeterDetailsByParam()

6.1.6 Oracle Procedures

Operation	Procedure Name
Load All Meters	V_SP_DA_ANALYZER_LOAD
Search Meters	V_SP_DA_ANALYZER_METER_SEARCH

6.1.7 Navigation

When a user clicks a meter number:

```
Analyzer Meter Grid
↓
openInNewTab()
↓
/meter-analysis-details/{meterNo}
```

6.2 Meter Analysis Details

Component	Files
MeterAnalysisDetailsComponent	meter-analysis-details.component.ts, meter-analysis-details.component.html

6.2.1 Tabs

- Detailed Demand
- Detailed Consumption
- Voltage Analysis

6.2.2 Shared Graph (Snowflake Chart Data) Architecture

All Meter Analysis charts use the same underlying architecture.

```
MeterAnalysisDetailsComponent
↓
MeterAnalyzerService
↓
GraphController
↓
GraphService
↓
Oracle Query Repository
↓
Snowflake Query
↓
Snowflake Database
↓
JSON Result
↓
Chart Rendering
```

6.2.3 Graph Controller and Service

Item	Detail
Controller	GraphController
Primary Endpoint	POST /Graph/GetSFGraphMeterDetail
Method	GetCombinedMeterData()
Service	GraphService
Key Methods	GetCombinedMeterData(), GetQueryFromOracle(), ReplacePlaceholdersFromJson()

6.3 Detailed Demand

6.3.1 Purpose

Displays demand-related metrics for a meter.

6.3.2 Supported Metrics

- kW Delivered
- kVA Delivered
- kW Received
- kVA Received
- Meter Temp (°C)
- Ambient Temp (°F)

6.3.3 Data Flow

Generate Graph Flow

```
Generate Graph
↓
callDetailedDemandMetricChartFunction()
↓
getMeterMetricDemandChartData()
↓
MeterMetricDataById()
↓
GraphController
↓
Query ID: 16
```

6.3.4 Channel Mapping

Channel	Description
1	kW Delivered
2	kVA Delivered
3	kW Received
4	kVA Received
8	Meter Temp
9	Ambient Temp

6.3.5 Demand Chart Series Mapping

Series	Metric	Channel
Series 1	kW Delivered	1

Series	Metric	Channel
Series 2	kVA Delivered	2
Series 3	kW Received	3
Series 4	kVA Received	4
Series 5	Meter Temp (°C)	8
Series 6	Ambient Temp (°F)	9

6.3.6 Demand Summary Calculations

Generated by: buildAllDemandSummaries(). For each selected unit, the following is computed: Start Date, Start Time, End Date, End Time, Maximum Value, Maximum Value Date, and Total.

Metric	Calculation
Total	Sum of all interval values
Maximum Value	Highest interval value
Maximum Value Date	Timestamp where the maximum occurred

6.3.7 Demand Calculation Functions

Function	Purpose
ConfigureDemandAxes()	Determine chart axis ranges dynamically
buildAllDemandSummaries()	Build the right-hand information panel
buildDemandInfo()	Calculate Start Date, End Date, Start Time, End Time, Maximum Value, Maximum Value Date, and Total

6.4 Detailed Consumption

6.4.1 Purpose

Displays consumption metrics.

6.4.2 Supported Metrics

- kW Delivered
- kVA Delivered
- kW Received
- kVA Received
- Meter Temp (°C)
- Ambient Temp (°F)

6.4.3 User Parameters

- Unit of Measure
- Unit of Time
- Ambient Temperature Toggle

6.4.4 Data Flow

Generate Graph Flow

```
Generate Graph
↓
callDetailedConsumptionMetricChartFunction()
↓
getMeterMetricConsumptionChartData()
↓
GraphController
↓
Query ID: 16
```

6.4.5 Unit of Time Options

- Year
- Month
- Day

Selected through: unitOfTimeValue

6.4.6 Consumption Summary Calculations

Generated by: buildAllConsumptionSummaries(). Calculates: Start Date, End Date, Start Time, End Time, Max Value, Max Value Date, and Total Consumption.

6.4.7 Consumption Calculation Functions

Function	Purpose

Function	Purpose
ConfigureConsumptionAxes()	Configure the consumption chart axis
buildAllConsumptionSummaries()	Build consumption summary panels
buildDemandInfo()	Calculate Maximum Value, Maximum Value Date, and Total Consumption

6.5 Voltage Analysis

6.5.1 Purpose

Displays voltage information for a meter.

6.5.2 Data Flow

Generate Graph Flow

```
Generate Graph
↓
callMeterMetricChartFunction()
↓
getMeterMetricVoltageChartData()
↓
GraphController
↓
Query ID: 15
```

6.5.3 Channel Mapping

Channel	Description
5	Phase A
6	Phase B
7	Phase C

6.5.4 Sidebar Calculations

Generated by: buildVoltagePhaseInfo(). For each phase (Phase A, Phase B, Phase C), the following is calculated: Start Date, End Date, Start Time, End Time, Maximum Value, Maximum Value Date, Minimum Value, and Minimum Value Date.

6.5.5 Voltage Calculation Functions

Function	Purpose
ConfigureVoltageAxes()	Dynamically split voltage lines onto separate axis groups
buildVoltagePhaseInfo()	Calculate Minimum Voltage, Maximum Voltage, Min Date, Max Date, Start Date, and End Date

6.6 Billing Period Data

Used whenever Billing Period is selected.

Item	Detail
Method	getBillingDateRangeData()
Query ID	17
Returned Fields	BEG_RD_DT, END_RD_DT, BILL_MONTH
Usage	Populate the Billing Period dropdown

Selecting a billing period automatically triggers:

- Demand Chart Load
- Consumption Chart Load
- Voltage Chart Load

7. Coincident / Non-Coincident Peak Reporting

Item	Detail
Component	AnalyzerComponent
Tab	Coincident/Non-Coincident Peak Reporting

7.1 Transformer Search

7.1.1 Search Criteria

- Point ID
- Pole/Pad ID
- Substation
- Circuit ID
- Crew Quarter

7.1.2 Data Flow

Transformer Search Flow

```
Search Component
↓
handleSearch()
↓
getAnalyzerTransformersDetailsBySearchParam()
↓
TransformerService
↓
POST AnalyzerPeakReportSearchData
↓
MeterAnalyzerController
↓
AnalyzerCoinNonCoinPeakReportSearch()
↓
V_SP_DA_COIN_NONCOIN_TRANSFORMER_SEARCH
↓
Oracle
↓
Grid
```

7.1.3 API Reference

Operation	Method	Endpoint
Load Transformers	GET	/MeterAnalyzer/AnalyzerPeakReportLoadAll
Search Transformers	POST	/MeterAnalyzer/AnalyzerPeakReportSearchData

7.1.4 Oracle Procedures

Operation	Procedure Name
Load All	V_SP_DA_COINC_NONCOIN_REPORTING_LOAD_ALL
Search	V_SP_DA_COIN_NONCOIN_TRANSFORMER_SEARCH

7.2 Coincident Peak Report Popup

Opened when a user clicks a Point ID.

Peak Report Popup Flow

```
User clicks Point ID
↓
openPointIdDialog()
↓
getTransformerCoinc_NonCoincData()
↓
transformerCoin_NonCoinData()
↓
Snowflake Query
↓
Contributor Meter Data
↓
Popup Grid
```

7.3 Coincident / Non-Coincident Query IDs

Query ID	Purpose
63	Coincident/Non-Coincident Query — 15 Minutes
65	Coincident/Non-Coincident Query — Hourly
66	Coincident/Non-Coincident Query — Daily Aggregated

7.4 Peak Calculations

Frontend calculates totals. Method: getTransformerCoinc_NonCoincData()

7.4.1 Totals

- coincidentKVAH
- coincidentKWH
- nonCoincidentKVAH
- nonCoincidentKWH

7.4.2 Calculation

Sum all contributor meter values for: COINCIDENT_KVAH_NET, COINCIDENT_KWH_NET, NONCOINCIDENT_KVAH_NET, NONCOINCIDENT_KWH_NET.

7.5 Popup Data Model

7.5.1 Header Values

- POINT_ID
- RUN_TIME
- UNIT_OF_MEASURE
- START_TIME
- END_TIME
- COINCIDENT_PEAK_TIME
- TOTAL_NON_COINCIDENT_PEAK

7.5.2 Grid Values

- SERVICE_POINT_ID
- METER_NO
- COINCIDENT_KWH_NET
- COINCIDENT_KVAH_NET
- NONCOINCIDENT_KWH_NET
- NONCOINCIDENT_KVAH_NET
- NONCOINCIDENT_PEAK_TIME

8. Export Functionality

8.1 Supported Exports

- Meter Search Grid
- Coincident/Non-Coincident Detail Popup

8.2 Implementation

Item	Detail
Method	exportGridByIndex()
Mechanism	Fusion Data Grid Excel Export

9. Calculation Framework

This section consolidates all business calculation logic referenced throughout the module.

9.1 Demand Calculations

Function	Purpose
ConfigureDemandAxes()	Determine chart axis ranges dynamically
buildAllDemandSummaries()	Build the demand information panel
buildDemandInfo()	Calculate Start/End Date-Time, Maximum Value, Maximum Value Date, Total

9.2 Consumption Calculations

Function	Purpose
ConfigureConsumptionAxes()	Configure the consumption chart axis
buildAllConsumptionSummaries()	Build consumption summary panels
buildDemandInfo()	Calculate Maximum Value, Maximum Value Date, Total Consumption

9.3 Voltage Calculations

Function	Purpose
ConfigureVoltageAxes()	Split voltage lines onto separate axis groups
buildVoltagePhaseInfo()	Calculate Minimum/Maximum Voltage and associated dates

9.4 Peak Calculations

Method: getTransformerCoinc_NonCoincData() — sums contributor meter values into COINCIDENT_KWH_NET, COINCIDENT_KVAH_NET, NONCOINCIDENT_KWH_NET, and NONCOINCIDENT_KVAH_NET.

? **Support Tip:** Incorrect totals should first be diagnosed at the frontend aggregation layer (buildAllDemandSummaries(), buildAllConsumptionSummaries(), and the coincident peak summation logic) before Oracle or Snowflake data is suspected.

10. Data Models and Request Objects

10.1 SearchAnalyzerMeterDetailsRequest

Field	Description
Meter_no	Meter number search parameter
Address	Address search parameter
City	City search parameter
Account_no	Account number search parameter
Point_id	Point ID search parameter

10.2 AnalyzerReportSearch

Field	Description
STATUS	Transformer/report status
POINT_ID	Transformer point identifier
POLE_PAD_ID	Pole/Pad identifier
SUBSTATION	Substation identifier
CIRCUIT_ID	Circuit identifier
CREW_QUARTER	Crew quarter identifier
DURATION	Reporting duration

11. Code Location Reference

11.1 Frontend Files

Component	Files
AnalyzerComponent	analyzer.component.ts, analyzer.component.html
MeterAnalysisDetailsComponent	meter-analysis-details.component.ts, meter-analysis-details.component.html

11.2 Backend Files

Layer	Files
Controllers	MeterAnalyzerController.cs, GraphController.cs
Services	MeterAnalyzerService.cs, GraphService.cs
Connectors	IOracleConnectorODP, SnowflakeConnector.cs

11.3 Oracle Stored Procedures Inventory

Procedure Name	Purpose	Called By
V_SP_DA_ANALYZER_LOAD	Load all meters	MeterAnalyzerController.AnalyzerAllMeterDetails()
V_SP_DA_ANALYZER_METER_SEARCH	Search meters by criteria	MeterAnalyzerController.SearchAnalyzerMeterDetails()
V_SP_DA_COIN_NONCOIN_TRANSFORMER_SEARCH	Search transformers by criteria	MeterAnalyzerController.AnalyzerCoinNonCoinPeakReportSearch()
V_SP_DA_COINC_NONCOIN_REPORTING_LOAD_ALL	Load all transformers	MeterAnalyzerController (Load Transformers)
V_SP_DA_GET_ALGQUERY	Retrieve Snowflake SQL template by Query ID	GraphService.GetQueryFromOracle()

12. Technology Stack and Known Dependencies

Layer	Technology
Frontend Framework	Angular
UI Framework	Fusion UI Components
Backend Framework	ASP.NET Core
Database	Oracle
Analytics Platform	Snowflake
Authentication	Azure AD / Enterprise Authentication
Charting	Fusion Chart Components

13. Troubleshooting Framework

The following table lists common failure scenarios, the components involved, and the recommended diagnostic checks.

Issue	Related Component(s)	Recommended Checks
Meter Search Returns No Data	AnalyzerAllMeterDetails API, V_SP_DA_ANALYZER_LOAD	Verify API response; verify Oracle connectivity; verify stored procedure execution
Meter Analysis Charts Empty	GraphController, Oracle Query Repository, Snowflake	Verify Query ID mapping; verify Oracle Query Repository entry; verify Snowflake query result
Billing Period Dropdown Empty	BillingDateRangeDataById(), Query ID 17	Verify Query ID 17 configuration; verify Snowflake query result
Coincident / Non-Coincident Popup Empty	Transformer Point ID, Query IDs 63/65/66	Verify Point ID validity; verify Snowflake result for the relevant Query ID
Incorrect Totals	buildAllDemandSummaries(), buildAllConsumptionSummaries(), Coincident Peak Summation Logic	Verify frontend aggregation logic before escalating to data layer

Important Note: Always confirm the Query ID mapping (Section 5.4) before escalating a blank-chart issue to the data platform team — most blank-chart issues trace back to a Query ID or Oracle Query Repository misconfiguration rather than a Snowflake data problem.

14. Technical Support

14.1 Support Approach

Support engineers should use this document as the primary reference when triaging Analyzer module issues. The recommended approach is:

- Identify the affected feature (Section 6 or Section 7).
- Confirm the associated Query ID using the Feature-to-Query Mapping (Section 5.3).
- Trace the data flow for that feature using the corresponding flowchart.
- Apply the relevant checks from the Troubleshooting Framework (Section 13).
- Escalate to the appropriate layer owner (Frontend, Backend/Service, Oracle, or Snowflake) only after the above steps have been completed.

14.2 Escalation Layers

Layer	Owning Area	Representative Artifacts
Frontend	Angular / UI Team	AnalyzerComponent, MeterAnalysisDetailsComponent, SearchComponent
Backend / Service	Application Development Team	MeterAnalyzerController, GraphController, MeterAnalyzerService, GraphService
Data — Oracle	Database Team	Stored procedures listed in Section 11.3
Data — Snowflake	Analytics Platform Team	Query templates resolved via V_SP_DA_GET_ALGQUERY

14.3 Document Maintenance

This document should be updated whenever a new feature, endpoint, stored procedure, or Query ID is introduced to the Analyzer module. Section numbering and the Table of Contents should be refreshed after any structural change.